NATIONAL TEACHERS COLLEGE

TITLE PROPOSAL

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Proposed Project Title 2: StudentShield: Machine Learning Framework for Student Bullying Detection & Automated Parent SMS Alerts for Junior High School at Saint Vincent Blessed School of Manila	

1.0 Area of Investigation:

The investigation centers on the design and development of **StudentShield**, a machine learning-powered web platform for Junior High School at SVBSM. This system enables anonymous venting of frustrations and depression symptoms, tracks bullying incidents (as victim or perpetrator), applies predictive behavioral analytics, and delivers automated SMS notifications to parents for proactive intervention.

2.0 Background and Rationale of the Project/System:

Bullying and student depression are pressing issues in junior high schools in SVBSM, often leading to poor mental health, academic decline, and disrupted learning environments. They rely on manual reporting and delayed counselor intervention, which fail to address patterns early or involve parents effectively.

StudentShield tackles these challenges through an integrated ML framework that analyzes vent entries for depression risks, detects bullying patterns via behavioral tracking, and triggers real-time SMS alerts to parents, fostering immediate discipline for perpetrators and support for victims. This creates a proactive, data-driven safety net that enhances student well-being, reduces incidents, and promotes a healthier school climate tailored to SVBSM's Junior High School student needs.

3.0 Statement of Objectives:

The main objectives of the **StudentShield** project are:

1. **Develop a secure web platform** for Junior High School students at SVBSM to anonymously vent frustrations, report depression symptoms, and log bullying incidents (as victim or perpetrator.)
2. **Implement machine learning models for predictive behavioral analytics** to detect patterns in depression risks and repeated bullying encounters from vent entries and tracking data.
3. **Integrate an automated SMS notification system** to alert parents immediately when the ML model calculates a high-risk threshold, enabling rapid home support while allowing counselors to monitor the data and follow up with the student afterward.
4. **Provide admin dashboards** for school counselors and teachers to monitor trends, review high-risk cases, and intervene proactively.
5. **Ensure data privacy compliance and a user-friendly, responsive web interface** tailored to junior high school students.

4.0 Scope and Limitations of the Study:

Scope:

- The study focuses on designing and developing the **StudentShield** web platform using PHP CodeIgniter, including venting forms, bullying trackers, ML sentiment analysis integration, and Twilio SMS automation.
- It will cover full implementation for SVBSM Junior High School students with admin oversight features.
- Pilot testing will validate detection accuracy and parent notification impact on behavior.

Limitations:

- The system targets only SVBSM Junior High School initially, with testing in a controlled school environment before broader rollout.
- ML models will use basic sentiment classification and pattern detection rather than advanced deep learning. SMS functionality depends on Twilio service availability and parent phone number accuracy.
- The study excludes long-term psychological impact evaluation and physical hardware integration, focusing solely on software functionality.

5.0 Target Beneficiaries:

1. **Junior High School Students at SVBSM:** StudentShield offers a safe space to anonymously vent frustrations and depression symptoms, reducing isolation while tracking bullying encounters to protect victims and encourage better behavior in perpetrators.
2. **Parents/Guardians:** Receive automated SMS alerts about incidents, enabling immediate home discussions, discipline for bullies, and support for victims to prevent escalation.
3. **School Counselors:** Gain admin dashboards with ML-driven insights on behavioral trends, high-risk students, and incident patterns for proactive interventions and reduced classroom disruptions.
4. **SVBSM Administration:** Benefits from a data-backed system that lowers bullying/depression rates, improves overall student well-being, and demonstrates commitment to modern mental health solutions.

6.0 References:

- McCloud, T., Jones, R., Lewis, G., Bell, V., & Tsakanikos, E. (2020). Effectiveness of a mobile app intervention for anxiety and depression symptoms in university students: randomized controlled trial. *JMIR Mhealth and Uhealth*, *8*(7), e15418. https://doi.org/10.2196/15418
- Madububambachu, U., Ukpebor, A., & Ihezue, U. (2024). Machine Learning Techniques to Predict Mental Health Diagnoses: A Systematic Literature Review. *Clinical Practice and Epidemiology in Mental Health*, *20*(1), e17450179315688. https://doi.org/10.2174/0117450179315688240607052117
- Ukpebor, Augustine. "Machine Learning Techniques to Predict Mental Health Diagnoses: A Systematic Literature Review." *Clinical Practice and Epidemiology in Mental Health*, vol. 20, no. 1, 26 July 2024, clinical-practice-and-epidemiology-in-mental-health.com/VOLUME/20/ELOCATOR/e17450179315688/. Accessed 1 Mar. 2026.
- Satpute, S. G., & Rajeshwar, G. T. (2025). Real-time AI solutions for monitoring and preventing bullying in Educational Institutions. *International Journal of Advanced Scientific Research*, 48–51. https://www.allscientificjournal.com/assets/archives/2025/vol10issue3/10052.pdf